

Product Requirements Document (PRD)

Recovery Pulse

A Zero-Typing GenAI Recovery & Relapse Prevention Companion for Individuals with Substance Use Disorders

1. Product Overview

Product Name

Recovery Pulse

Elevator Pitch

Recovery Pulse is a **zero-typing, AI-powered emotional check-in companion** that helps individuals recovering from Substance Use Disorders communicate how they feel in under five seconds. Using simple emoji-based check-ins or voice input, the platform identifies emotional distress, estimates relapse risk, provides immediate AI support, and keeps caregivers informed when necessary.

2. Problem Statement

Individuals recovering from substance use disorders spend over **99% of their recovery journey outside clinical settings**.

During this time they often experience:

- Cravings
- Stress
- Loneliness
- Anxiety
- Depression

- Emotional triggers

Unfortunately,

- Therapists are unavailable.
- Caregivers don't know something is wrong.
- Users hesitate to type long messages or ask for help.

As a result, many relapses occur without early intervention.

3. Goal

Reduce relapse risk by making emotional check-ins effortless.

One tap. One voice message. Immediate support.

4. Target Users

Primary

People recovering from:

- Alcohol addiction
 - Drug addiction
 - Nicotine dependence
 - Prescription medication misuse
-

Secondary

- Parents
- Partners
- Caregivers
- Recovery coaches
- Counselors

5. Success Metrics

User Metrics

- Daily check-in completion rate
- Consecutive recovery streak
- Voice journal usage
- Number of SOS activations

Clinical Metrics

- High-risk emotions detected
- Early interventions triggered
- Reduction in missed daily check-ins

Business Metrics

- Daily Active Users
 - Weekly Retention
 - Caregiver engagement
-

6. User Journey

Scenario

Rahul completed rehabilitation last month.

At 8 PM his phone sends:

"How are you feeling today?"

Rahul taps

 Craving

Within seconds,

Recovery Pulse

- detects elevated relapse risk
- responds empathetically
- recommends coping strategies
- offers caregiver notification
- records today's emotional state

No typing required.

7. Core Features

Feature 1

One-Tap Mood Check-in

User selects one option.

Emoji	Meaning
	Feeling Good
	Okay
	Struggling
	Craving

Time required

<3 seconds

Feature 2

Hold-to-Talk

Large microphone button.

User speaks naturally.

Example

"Today has been really difficult."

GenAI

- converts speech to text
- analyzes emotion
- detects craving signals
- generates supportive response

No typing.

Feature 3

AI Recovery Companion

Immediately after every check-in

AI responds like a recovery coach.

Example

"I'm proud that you checked in today. Cravings are temporary. Let's work through this together."

Feature 4

Relapse Risk Score

Generated using

- mood
- voice sentiment
- previous check-ins

- streak

Risk Levels

 Low

 Medium

 High

Feature 5

AI Suggested Action

Depending on risk.

Examples

Low

Take a short walk.

Medium

Try a 3-minute breathing exercise.

High

Contact your support person immediately.

Feature 6

Caregiver Notification (Optional)

Only after user consent.

Example notification

Rahul reported feeling overwhelmed today. A supportive check-in may help.

No medical diagnosis.

Only wellness information.

Feature 7

Recovery Streak

Daily check-ins increase streak.

Example

🔥 12 Day Recovery Pulse

Positive reinforcement.

8. Functional Requirements

Authentication

- Email
 - Google Login
-

Home Screen

Display

Recovery Pulse

"How are you feeling?"

Emoji buttons

Microphone button

Recovery streak

Mood Check

Save

- timestamp
 - mood
 - user ID
-

Voice Journal

Accept

30–60 second voice message.

Generate

- transcript
 - emotion score
 - AI response
-

AI Engine

Input

Mood

Voice transcript

Previous history

Output

- supportive message
 - relapse risk
 - recommended action
-

Notification Engine

Daily reminder

Customizable time.

Example

8 PM

Caregiver Dashboard

View

- latest mood
 - streak
 - alerts
-

9. Non-Functional Requirements

Response time

<2 seconds

Availability

99%

Mobile responsive

Yes

Accessibility

Large buttons

Voice-first interaction

High contrast mode

10. Tech Stack

Frontend

- React
- TailwindCSS

Backend

- Django REST Framework

Database

- PostgreSQL

Speech Recognition

- Whisper API

LLM

- OpenAI GPT

Authentication

- Firebase Auth

Notifications

- Firebase Cloud Messaging
-

11. AI Prompt Example

Input

Mood: Craving

Previous Mood:

Good

Good

Struggling

Voice:

"I don't think I can handle today."

Prompt

You are an empathetic recovery coach.

Respond in less than 100 words.

Do not shame.

Encourage healthy coping.

Suggest one actionable next step.

If severe emotional distress is detected,
recommend contacting a trusted person.

Tone:

Warm

Supportive

Non-judgmental

12. Screens

Screen 1

Splash

Recovery Pulse

Screen 2

Home

Recovery Pulse

How are you feeling today?



 Hold to Talk

 12 Day Streak

Screen 3

AI Response

I'm really glad you checked in today.

Cravings don't define your recovery.

Let's focus on getting through the next
10 minutes together.

Suggested Action

✓ Drink water

✓ Deep breathing

✓ Call support person

[Notify Caregiver]

Screen 4

Caregiver Dashboard

Rahul

Today's Mood



Risk

Medium

Last Check-in

8:03 PM

Recovery Streak